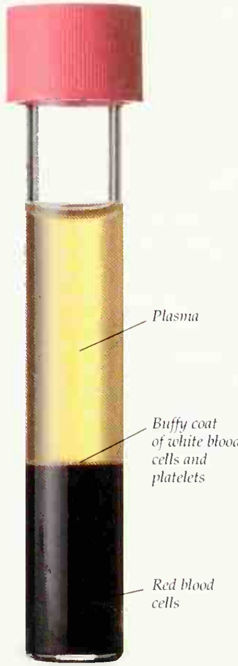
Oxygenated blood

Deoxygenated blood

WHY IS BLOOD RED?
Dutch microscopist Antonie van Leeuwenhoek proposed that blood's redness is due to the "small globules" we now call red cells, though he would have seen them as pale yellow blobs through his magnifying lens. Each red cell contains a protein called hemoglobin, which is an oxygen transporter. Hemoglobin happens to be bright red when combined with oxygen and a darker bluish red when the oxygen has been given up.

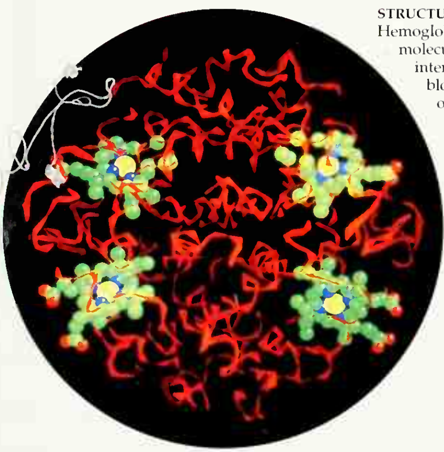
WHAT IS IN BLOOD?
Allow a blood sample to settle, and it reveals its main components. Just over half (55 percent) is a pale fluid, plasma. This is more than nine-tenths water and contains dissolved sugars, salts, wastes, body proteins, hormones (p. 40), and many other chemicals. The thin layer, called the buffy coat, is disease-fighting white cells and blood-clotting platelets. The rest, about 45 percent, is made up of red cells.



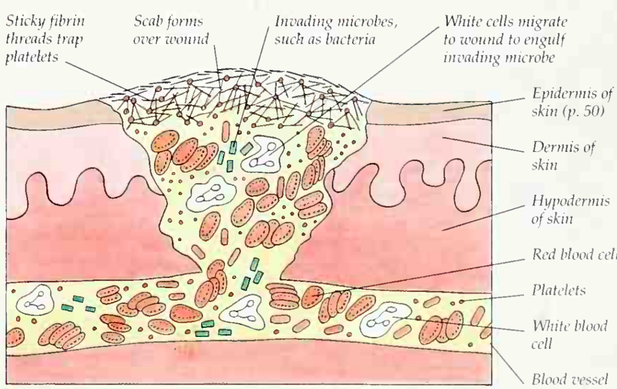
Settled blood

RED BLOOD CELLS
Also known as erythrocytes or red corpuscles, red blood cells are some of the smallest cells in the body, 0.0003 in (0.007 mm) wide. They are concave on each side. Red cells are made in the bone marrow (p. 17), where they lose their nuclei (control centers), then live for three to four months in the blood.

WHITE BLOOD CELLS
Also known as leukocytes or white corpuscles, white blood cells are not really white, but ghostly pale, jellylike, and able to ooze and change shape. There are several types and sizes. Their main role is to clean the blood, consume debris, and battle against invading viruses and other microbes. To search out invaders, they can squeeze between the cells forming the capillary walls and wander into the surrounding tissues with flowing movements.



STRUCTURE OF HEMOGLOBIN
Hemoglobin, shown here as a computer-generated molecular model, is a protein made from four intertwined chains of amino acid building blocks, with about 10,000 atoms in total. Four of these are iron atoms (yellow), cradled in four heme rings (green) of amino acids, which act like oxygen magnets. In the high-oxygen environment of the lungs, each heme group picks up a linked pair of oxygen atoms. In the lower-oxygen tissues, the oxygen atoms detach and drift away for cellular respiration (p. 24). There are around 300 million molecules of hemoglobin in each red cell.



FORMING BLOOD CLOTS
In a pinprick of blood there are about 5 million red cells, about 10,000 white cells, and 250,000 platelets, which are also known as thrombocytes. The platelets are cell fragments from the bone marrow. Damage to a blood vessel begins a cascade of chemical reactions in which a dissolved blood protein, fibrinogen, changes into insoluble threads of fibrin. These form a crisscross network. Platelets stick to the vessel's walls and the fibrin mesh. Red cells get entangled, too, and the whole barrier becomes a clot that seals the leak.